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GAUGE: When Not to Trust LLM-as-a-Judge in User-Simulated Evaluation of Task-Oriented Agents

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Abstract

Comparing and selecting task-oriented LLM agents increasingly relies on a low-cost offline evaluation gate: persona-driven LLM user-simulators converse with each candidate, an *LLM-as-a-judge* scores the transcripts, and the higher-scoring agent is promoted. We introduce GAUGE, a reusable offline protocol that measures whether this gate's *ranking* matches a grounded *verifiable reward* across 25 agents from six providers on the τ^2-bench and SimulatorArena benchmarks, separating two kinds of evaluation validity that release practices conflate: *ranking validity* and *construct validity*. First, a **satisfaction–success gap**: satisfaction carries essentially no information about task success, as conversations rated *satisfied* by our blind panel are decorrelated from actual success, with 57.5% of them failing the customer's task, a pattern consistent across five rater populations, both benchmarks, and every subjective dimension we rated. Second, while the gate's *ranking* is robust across the broad capability span, it loses resolution among the near-equal strong agents: this *decision-disagreement rate* jumps from $<1\%$ on wide-reward pairs to 31% on close pairs. The gate is thus *human-validated yet mis-anchored*. As a remedy, we propose a *calibrate-then-trust* cadence in which a judge-free completion bit is a zero-cost tripwire for *truncation* regressions.

1 Introduction

Developers and researchers building agentic, task-oriented LLM agents for customer service, tool-use assistants, and tutoring must decide which variant is better, under three constraints: there is no clean held-out test set that captures live user experience, a controlled human study per candidate config is prohibitively slow, and the config space (model, prompt, policy, tools) changes often. The field's de facto answer is a low-cost offline gate: drive each candidate against persona-conditioned LLM user-simulators, score the transcripts with an LLM-judge, and promote the higher-scoring variant. It runs in continuous integration (CI) at a few cents per transcript with no labeled production data, resting on one assumption teams almost never measure: that this composite simulator-plus-judge gate ranks variants the way a grounded evaluation would. We audit this gate as deployed and pair the audit with a drop-in zero-cost mitigation. That LLM judges exhibit systematic biases, including self-preference and position bias, is known (Panickssery et al., 2024; Wang et al., 2024); what has not been quantified is their magnitude at the release-decision unit against a verifiable reward: a 57.5% false-accept rate on the blind human panel, a gap that persists across five rater populations and alters the resulting agent-selection decisions.

	SimArena/ LostInSim	ECom-Bench	Judge-eval	Chat-Bench	**GAUGE**
Ranking validity vs. reward	∼	∼	✗	∼	✓
Release-decision unit	∼	✓	✗	∼	✓
Satisfaction ≠ task success	✗	✗	✗	∼	✓
Cross-provider scale	∼	✗	∼	✗	✓
Grounded in real human ratings	✓	✗	✓	✓	✓
Judge-free cost decomposition	✗	✗	✗	✗	✓

Table 1: Positioning against prior evaluation work. Only GAUGE tests whether the simulator+judge *gate* ranks agents like a verifiable non-LLM reward and quantifies the satisfaction–success gap. ✓ fully addresses the criterion; ∼ partial; ✗ not.

We introduce **GAUGE** (Grounded Audit of User-simulator-and-judge Gate Evaluation), a reusable, fully offline protocol that separates two kinds of validity that release practice conflates (Table 1). The gate has *ranking validity* (it orders agents like the verifiable reward) and is certified for *construct validity* against human *satisfaction*; yet satisfaction is not success (the *satisfaction–success gap*): 57.5%

of conversations our blind human panel rated *satisfied* ($\geq$5/7) had failed the customer's task, a gap that misleads the LLM-judge and the humans. A gate can be human-validated yet mis-anchored.

The satisfaction anchor remains misleading as a release signal despite ranking validity, for three reasons. First, the ranking holds only where capability dominates: among the *near-equal* strong agents real release decisions compare, the gate promotes the lower-reward agent on 31% of close pairs (the *decision-disagreement rate*; §4.2), up from $<$1% where rewards are far apart. Second, teams optimize against gates rather than only reading them, and tuning to a satisfaction anchor raises a metric known to diverge from the objective once optimized (Strathern, 1997; Skalse et al., 2022). Third, an accept/reject threshold anchored on satisfaction admits agents that fail 48–60% of the time (§4.1). The anchor thus governs exactly the decisions a gate exists to make: promoting near-equal candidates, serving as an optimization target, and setting the absolute accept/reject bar. The user-simulator is a component of the gate we audit: every central claim is anchored on the simulator-independent verifiable reward and human panel.

GAUGE operates as a *periodic calibration* (a *calibrate-then-trust* cadence) rather than a per-decision gate: the verifiable audit is run once on a representative benchmark to learn the gate's trusted operating region, after which the cheap gate is used in CI within that region. It applies wherever an offline verifiable reward is obtainable, here the τ^2-bench oracle: DB-state and action checks, and SimulatorArena's human-graded correctness. Our contributions are as follows.

(1) GAUGE, a reusable protocol for validating simulator-plus-judge evaluation. We introduce the first protocol to validate a simulator-plus-judge evaluation gate against a verifiable, non-LLM reward at the release-decision unit, applied at scale: 25 agents, six providers, four judges, two substrates, $\approx$3,700 transcripts (§3).

(2) Satisfaction–success gap. We show that satisfaction carries essentially no information about task success: 57.5% of *satisfied* conversations failed the task ($\rho = -0.147$), a gap that holds across five rater populations and two substrates (§4.1).

(3) Separating ranking from construct validity. We separate *ranking* from *construct* validity on the six-provider ladder: the ranking is robust (ρ=0.94) yet loses resolution among near-equal agents, where the gate promotes the lower-reward agent on 31% of close pairs, and we isolate same-family judge self-preference (+0.75/7; §4.2, §4.3).

(4) Calibrate-then-trust recipe. We show that a judge-free completion bit is a zero-cost tripwire for truncation regressions (ρ=0.87 vs. 0.80 on broken-vs-working; §4.5), and report a negative result: out-of-sample recalibration does not transfer (§4.6).

Key Findings.

- Subjective approval is decorrelated from task success across all dimensions, so an evaluation gate can be human-validated yet mis-anchored to the outcome.
- High aggregate ranking validity coexists with unreliable resolution of near-equal candidates, precisely the comparisons release decisions depend on.
- Evaluator reliability is determined by evidence access, not model capability: outcome-grounded judging succeeds where transcript-only satisfaction fails.
- The gate is valid only within a bounded operating region, requiring re-auditing on configuration change rather than on a fixed schedule.

2 Related Work

User-simulator reliability. Simulated users have long evaluated dialogue agents, from satisfaction-driven simulation (Sun et al., 2021) to LLM-based simulators (Davidson et al., 2023; Sekulic et al., 2024) and "agents evaluating agents" (Zhuge et al., 2025). SimulatorArena (Dou et al., 2025) and Lost-in-Simulation (Seshadri et al., 2026) test LLM user-simulators as human proxies via absolute rating correlation or single-agent reliability. *Delta:* we audit the composite gate's agent-ranking validity at the release-decision unit against a non-LLM verifiable reward, and show its optimized satisfaction signal is decorrelated from task success.

Agent and customer-agent benchmarks. Tool-using agents are evaluated by recent benchmarks (Zhou et al., 2024; Qin et al., 2024; Liu et al., 2024); ECom-Bench (Wang et al., 2025) reports persona-simulator pass-rates, while τ-bench (Yao et al., 2025) and its successor τ^2-bench (Barres et al., 2026) provide a verifiable substrate. *Delta:* we test whether such pass-rates rank agents like a grounded reward, and surface the satisfied-but-failed inversion.

LLM-as-a-judge calibration to humans. A large literature validates LLM-judges against human preference and satisfaction (Zheng et al., 2023; Liu et al., 2023; Gu et al., 2024), documents judge

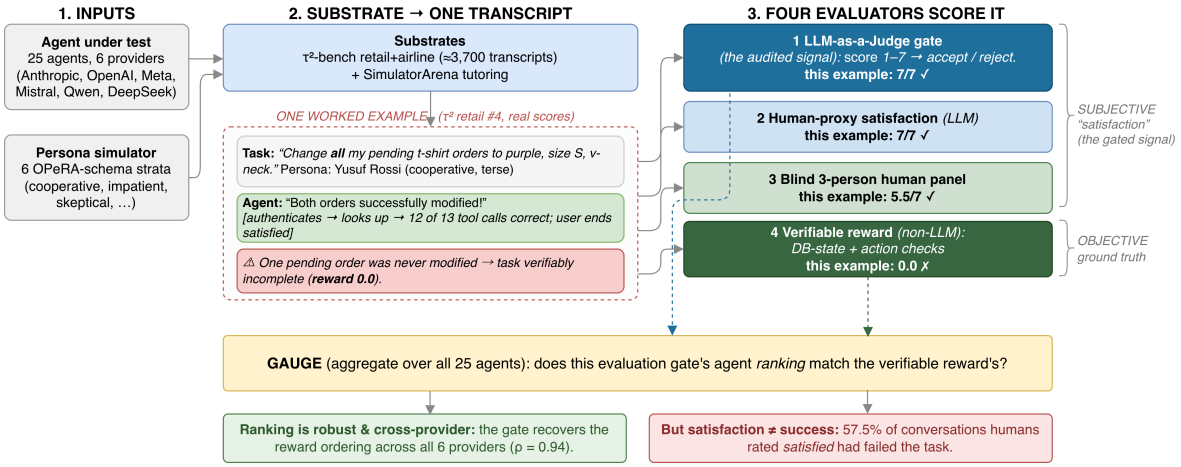


Figure 1: GAUGE measures whether the LLM-as-a-Judge and user-simulator evaluation gate ranks agents the way a grounded, verifiable reward would. **Left to right:** an agent and a persona simulator converse on a substrate, and four evaluators score each transcript: three subjective satisfaction signals (the LLM-as-a-Judge gate, an LLM human-proxy, and a blind 3-person human panel) and one objective non-LLM reward. Aggregating over all 25 agents, the gate's agent *ranking* matches the reward's (ρ=0.94), yet satisfaction is decorrelated from success.

biases (position (Wang et al., 2024), self-preference (Panickssery et al., 2024), inconsistency (Stureborg et al., 2024)) and benchmarks LLM judges and reward models (Tan et al., 2025; Lambert et al., 2025). A complementary line calibrates model confidence to signal when to trust an LLM output versus defer to a human (Bodhwani et al., 2025a; Jung et al., 2025). *Delta:* for customer agents this anchor is misaligned: human satisfaction is decorrelated from task success, so a judge can be "human-validated" yet mis-rank agents. Where recent work finds simulators over-cooperative or ranking-divergent (Mind-the-Sim2Real (Zhou et al., 2026), SimulatorArena (Dou et al., 2025)), we show the decorrelation misleads humans too.

Measurement validity and proxy gaming. Our framing draws on construct validity (Cronbach and Meehl, 1955) and its ML translation (Jacobs and Wallach, 2021; Bowman and Dahl, 2021): a metric can order systems well yet miss the construct it certifies, and an optimized proxy diverges from the objective (Goodhart's law (Strathern, 1997), reward hacking (Amodei et al., 2016; Skalse et al., 2022)). *Delta:* we give a measured instance for release gates, adapting controlled-perturbation instrument validation (Ribeiro et al., 2020; Adebayo et al., 2018) into our flag-only degradation set (§4.5, Appendix C). Unlike ChatBench (Chang et al., 2025), which closes the offline-vs-interactive gap by fine-tuning, GAUGE is fine-tuning-free.

3 Method

Figure 1 shows the audit end to end: an agent and a persona simulator converse on a substrate, and four evaluators score each transcript, one of them a verifiable, non-LLM reward.

Problem formulation Let $\mathcal{A}$ be a set of candidate agents, $\mathcal{P}$ persona strata, and $\mathcal{D}$ domains. For agent a, persona p, domain d, the simulator and agent produce a transcript $\tau_{a,p,d}$ carrying a per-transcript verifiable correctness label $r(\tau) \in \{0, 1\}$ from a non-LLM oracle. Three subjective signals also score the transcript on a satisfaction scale: an LLM-judge *gate*, an LLM *human-proxy*, and a *human panel*; write any such signal as $s(\tau)$. At the release-decision unit (the agent a; we aggregate over p, d, since per-cell scores are too noisy to act on) we define the agent-level gate $G(a) = \mathbb{E}_{p,d}[\text{gate}(\tau)]$ and verifiable reward $R(a) = \mathbb{E}_{p,d}[r(\tau)] \in [0, 1]$, and likewise $S_{\text{proxy}}(a), S_{\text{hum}}(a)$. GAUGE measures two quantities. *Ranking validity* $\rho(G, R)$ over $\mathcal{A}$ measures whether the gate orders agents like the verifiable reward. The *construct gap* (equivalently, the *satisfaction–success gap*) is the rate at which the task failed ($r(\tau)$=0) among transcripts a signal rated satisfied (a high $s(\tau)$: $\geq$5 on a 7-point or $\geq$8 on a 10-point scale). Crucially, a gate can be human-validated ($\rho(S_{\text{hum}}, G)$ high, the judge tracks human satisfaction) yet mis-anchored ($\rho(S_{\text{hum}}, R)$ low), because satisfaction itself does not track success.

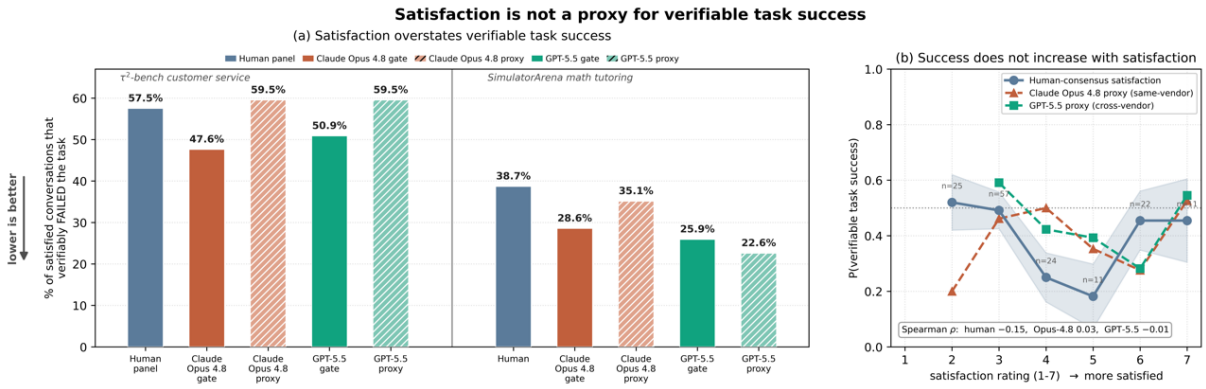


Figure 2: Satisfaction versus verifiable task success across raters. **(a)** Share of each rater's *satisfied* conversations that nonetheless failed the task, on the τ^2 human panel and the independent SimulatorArena substrate. **(b)** Empirical $P(\text{success})$ by satisfaction rating (integer bins, $n \geq 5$; human ± 1 SE shaded): success does not rise with satisfaction for any rater.

Substrates and ground truth. Our primary benchmark (substrate) is τ^2-bench (Barres et al., 2026) retail and airline, whose verifiable reward is a non-LLM oracle, fully deterministic on airline (DB-state and communicate checks) and predominantly deterministic on retail (a deterministic DB check gates an LLM-scored natural-language assertion; Appendix K); for replication we add SimulatorArena (Dou et al., 2025) math tutoring with human satisfaction ratings and human-graded correctness. Per-substrate statistics in Appendix J.

Agents. We evaluate two complementary sets. The *cross-provider ladder* (§4.2, §4.3) spans 14 models from six providers (Anthropic, Meta, Mistral, Qwen, DeepSeek, OpenAI) at two temperatures on the full task pools of 114 retail and 50 airline tasks, yielding 25 scored model-temperature configurations and about 3,700 transcripts in total (Appendix J). This wide capability span makes ranking validity, and its near-equal limit, measurable. The *controlled-degradation set* (§4.5, Appendix C) is a positive control: 12 Sonnet-4.5 configurations of *a-priori*-known good, medium, or degraded quality, crossed with 6 strata and 2 domains to give 144 cells and 720 transcripts, degraded only through inference-time hyperparameters such as temperature, token or step limits, and error tolerance, with no prompt or code edits. It supplies the broken-vs-working axis the all-strong grid cannot, so a signal that fails to rank the degraded configs last exhibits a measurable validity failure (Ribeiro et al., 2020; Adebayo et al., 2018).

Judges and humans. We score each transcript with two deliberately disjoint satisfaction rubrics (prompts in Appendix L). The *gate* is policy-aware: an operations-supervisor rubric that reads the full transcript (tool calls and goal included) and scores service quality with policy adherence and task resolution first. The *proxy* is satisfaction-only: a process-blind first-person shopper (tool calls and task stripped) rating how the conversation felt, standing in for the human rater gates are usually validated against. Disjoint in role, evidence, and vocabulary, the two cannot agree by a shared-rubric artifact. Four judges apply the gate rubric to the full grid (Claude Sonnet-4.5 and Opus-4.8, GPT-5.4 and GPT-5.5), with Opus-4.8 the primary judge (Anthropic, 2026b) and GPT-5.5 its out-of-family counterpart; results are consistent across all four (§4.3). A blind 3-person panel grounds the scores on a stratified 150-transcript τ^2 sample.

Metrics. We report agent-level Spearman correlation with model-clustered bootstrap CIs (Efron, 1979); broken-vs-working AUC; a power analysis (min-detectable ρ); out-of-sample recalibration (leave-one-stratum/agent-out); and panel reliability as Krippendorff's α (Krippendorff, 1980) against a human–human ceiling (Artstein and Poesio, 2008).

4 Results

4.1 Satisfaction does not imply task success

On a stratified 150-transcript τ^2 sample rated by three blind, independent annotators (a fully-crossed 150×3 design; Krippendorff's α=0.79, human–human ceiling ρ=0.85), two independently-built LLM judges reproduce human satisfaction (Opus-4.8 ρ=0.846, GPT-5.5 ρ=0.827). Yet satisfaction carries essentially no information about task suc-

Signal	P(fail | satisf.)	base rate	RR	AUC
Human satisfaction (panel)	57.5	57.3	1.00	0.44
Satisfaction proxy (grid)	32.7	40.2	0.81	0.49
Policy-aware gate (grid)	**20.0**	40.2	**0.50**	**0.73**

Table 2: Satisfaction matches the base failure rate; only the policy-aware gate improves on it. For each signal: the failure rate among the conversations it rated *satisfied*, the pool's own base failure rate, their ratio (relative risk, RR), and transcript-level discrimination (AUC). The human panel is a stratified sample (base 57.3%); the grid signals use the natural task mix (base 40.2%).

cess: across all 150 transcripts the correlation is flat and non-positive ($\rho = -0.147$; dose-response curve flat, Fig. 2b). Crucially, this is not an artifact of the word "satisfaction": all five subjective dimensions the panel rated (satisfaction, respect, clarity, perceived helpfulness, and would-return) are likewise decorrelated from success ($|\rho| \leq 0.17$; Appendix B), so the gap reflects subjective approval in general. Concretely, 57.5% of the conversations the panel rated *satisfied* ($\geq$5/7) had failed the task.

Uninformative, not merely weak. Interpreted against the base rate, this pattern is the central finding (Table 2). The 57.5% satisfied-but-failed rate is statistically indistinguishable from the stratified sample's own 57.3% base failure rate: conditioning on *satisfied* does not lower failure, so satisfaction is *uninformative* rather than merely weak, and at the transcript level it does not discriminate success (AUC 0.44). By contrast, the policy-aware gate assesses whether the task resolved: it roughly halves failure risk (20.0% of gate-accepted conversations fail against a 40.2% base rate on the natural task mix) and discriminates success (AUC 0.73; per-score calibration in Appendix C.1). The process-blind *proxy*, which never sees the tool calls or task, behaves like satisfaction, not the gate (32.7% vs. 40.2%; AUC 0.49). Separating these two signals, one uninformative and one informative, is the core contribution of this work. The panel result is robust to dropping any single annotator (56.4–60.5% leave-one-annotator-out; Appendix A).

Robust across raters and domains. The satisfied-but-failed rate is robust across sample scale, raters, and providers: it is concordant across five rater populations (the human panel plus gate and proxy scorings from two LLM providers; 47.6–59.5%; Fig. 2a, Table 4) and stable across satisfaction thresholds. It also holds within each domain: satisfied-but-failed is 24.1% (retail), 62.1% (airline), and 38.7% (math tutoring), and the ranking replicates per domain (ρ=0.95 retail, 0.80 airline; Appendices F, H). The process-blind proxy overstates success more than the policy-aware gate, which observes whether the task resolved.

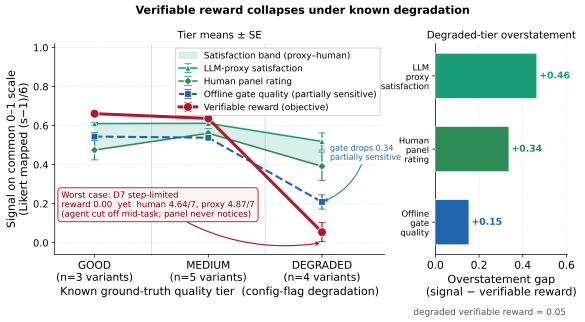


Figure 3: Construct-gap positive control: an agent degraded by configuration flags across three known-quality tiers, all signals on a common $[0, 1]$ scale. As quality drops good→degraded, verifiable reward collapses ($0.66 \rightarrow 0.05$) while the satisfaction signals stay nearly flat and the policy-aware gate is only partly sensitive ($0.54 \rightarrow 0.21$); the right panel quantifies each signal's overstatement at the degraded tier.

Not a simulator or reward-term artifact. The inversion surfaces only where satisfaction is the operative signal: where verifiable reward already separates agents cleanly, as on the strong-agent six-provider grid, the gate-accept inversion rate falls to 20%, the false-accept ceiling a construct-valid gate should meet (292/1,460, $p = 0.51$). As qualitative cross-substrate corroboration, it replicates on an independent, human-grounded substrate, SimulatorArena math tutoring, where 38.7% of conversations rated $\geq$8/10 by humans were verifiably incorrect ($p = 0.013$; Fig. 2a, right). Swapping our Sonnet-4.5 user-simulator with an independent simulator: GPT-5.4, with agent, tasks, rubrics, and oracle fixed, leaves the inversion intact (66.7% gate-satisfied and 69.8% proxy-satisfied still failing; Appendix D). On airline, whose reward is fully deterministic, the proxy still false-accepts 62.1% of *satisfied* conversations (325/523). A degradation positive control demonstrates this directly (Fig. 3): as one agent is degraded across known-quality tiers, reward collapses while satisfaction stays flat; the starkest case is a transcript truncated mid-task, scoring reward 0.0 yet earning the highest human rating of any degraded variant (4.6/7; Appendices C, M).

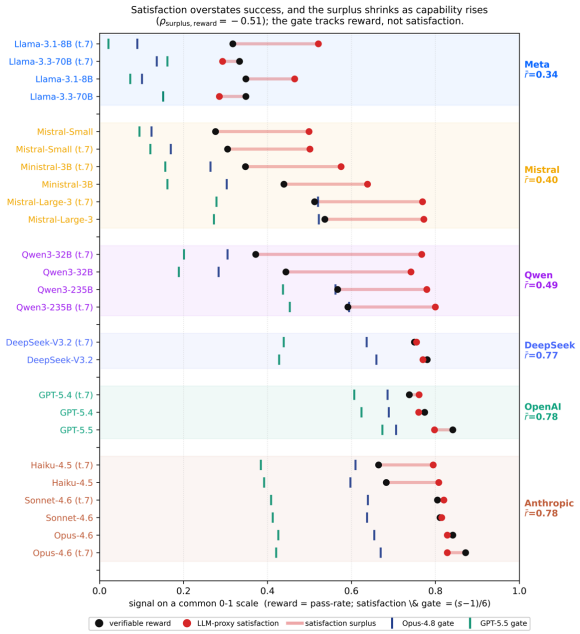


Figure 4: Per-agent verifiable reward, LLM-proxy satisfaction, and the two frontier-judge gates (Opus-4.8, GPT-5.5) for all 25 agents. The bar is the *satisfaction surplus* (satisfaction−reward). Both gate ticks track reward rather than satisfaction, and every agent is rated more satisfying than it succeeds.

4.2 Ranking is robust across providers

The construct gap might suggest the gate is invalid, yet its *ranking* is sound. Across the 25-agent, six-provider ladder (Fig. 4; per-agent values in Appendix E), the Opus-4.8 gate recovers the verifiable-reward ordering at ρ=0.94, far above the minimum correlation resolvable with N=25 agents (ρ=0.54 at $p<0.05$); the ranking holds across both domains and all four judges and cleanly separates broken from working agents on the controlled-degradation set (AUC=1.00; per-judge and per-domain coefficients in Appendix F). The ordering persists under an independent-provider user-simulator: rerunning the full 25-agent grid with GPT-5.4 (agent and oracle fixed) preserves it at ρ=0.93, against 0.94 under Sonnet-4.5, so the ranking reflects agent quality rather than same-family simulator–agent affinity (Appendix D).

This validity is driven by breadth: it lapses among the near-equal strong agents that real release decisions compare. Such pairs are common (30% of all pairs and 55% of top-half pairs differ in verifiable reward by less than 0.1, our near-equal threshold, near the oracle's resolution given per-agent sampling noise), so we report the *decision-disagreement rate*: how often the gate promotes the strictly *lower*-reward agent of a near-equal pair. The primary Opus-4.8 gate does so on 31% of them versus <1% on wide pairs, and the four other signals behave alike (GPT-5.4 30%, GPT-5.5 39%, human-proxy 30%, pooled gate 38%; Appendix F), over thirty times the 0.9% wide-pair rate. This separation is robust to resampling: the near-equal rate's base-model cluster-bootstrap 95% CI is $[11.6, 50.0]\%$, whose lower bound alone is $\sim 12\times$ the wide-pair rate. This is a resolution limit at deployment-relevant separations, not a non-significant correlation. GAUGE thus certifies the gate as a regression and coarse-quality detector and flags where it cannot adjudicate, a safety-relevant gap current gate-based evaluation overlooks.

4.3 Judge robustness and self-preference

Because the Anthropic judge and the agents share a provider, "validity" could reflect self-recognition (Panickssery et al., 2024), so we re-score the grid with an out-of-family judge. Opus-4.8 and GPT-5.5 agree at ρ=0.92, all four judges recover the verifiable-reward ordering (ρ=0.84–0.94) and agree with one another (ρ=0.90–0.98), and the ordering is thus not a single-provider artifact, though GPT-5.5 grades ≈0.8/7 lower.

The ordering is also not a capability artifact. Run as agents on the same τ^2 pool, both frontier judges are statistical peers of the strongest agent (Opus-4.8 0.82, GPT-5.5 0.84, vs. pool-best Opus-4.6 (t.7) at 0.87; overlapping 95% CIs); they recover the ordering without outclassing it, while the weakest judge-as-agent (GPT-5.4, 0.77) is tied-best as a ranker (ρ=0.94). The near-equal limit holds for every judge (top-11 ρ=0.25–0.51, all n.s.; Appendix F.1), confirming genuine agent near-equality rather than mis-ranking by a weak judge.

The two-provider design also isolates same-family self-preference: the Opus-4.8 judge inflates Claude agents over GPT-5.5 more than non-Claude agents, a scale-invariant diff-in-diff of $+0.75$/7 (Sonnet-4.5 shows $+0.67$/7 against GPT-5.4; Appendix F.2), so it reflects self-preference and leaves the ranking intact. Two reruns confirm the ranking is stable against judge noise (test–retest ICC=0.87, rank stability ρ=0.92; Appendix F.3).

4.4 Personas stress-test agents

The simulator drives each agent against six persona strata spanning cooperativeness, patience, and assertiveness (axes informed by the OPeRA persona schema (Wang et al., 2026); overlays in Ap-

Signal	$/dec.	ρ	Use / cadence
Completion bit	**0**	**0.87**	regressions; every CI run
Agent cost / length	0	0.17–0.34	–
LLM-judge gate	+0.60	0.80	coarse rank; bit insufficient
LLM-proxy sat.	+0.60	0.35	tone only, *not* success
Verifiable audit	rollouts	–	calibration; periodic

Table 3: Release signals by marginal $/decision, Spearman ρ vs. the verifiable reward (controlled-degradation set, N=12 variants), and recommended use/cadence. The zero-cost completion bit catches severe regressions as well as the paid judge (ρ 0.87 vs. 0.80); agent-cost and proxy ρ (0.17–0.35) are not significant.

pendix I). A single cooperative user masks failures: relative to the default τ^2 baseline (12 variants, retail; Fig. 10a), the persona population cuts mean verifiable reward by 42% (0.47 $\rightarrow$ 0.27).

The personas also expose the construct gap (Fig. 10b): satisfaction swings widely by stratum (proxy 3.5–5.4/7) yet does not track success (the anxious/low-trust and skeptical strata tie on reward, 0.38, but differ by 1.2/7), so a satisfaction-optimized signal tracks who the user is rather than whether the task succeeded. The benchmark stays stable: every degraded variant floors at reward 0 and all working variants stay above (whole-set reward-rank ρ=0.83), so the ranking validity (§4.2) reflects agent quality, not the simulator.

4.5 A free signal catches severe regressions

A judge-free *completion bit* (conversation completed), parsed at zero cost from existing rollouts, is a targeted tripwire. On the controlled-degradation set (broken-vs-working) it scores ρ=0.87 against the verifiable reward versus 0.80 for the full gate (Table 3), because those failures are truncation-style. But truncation is the minority failure mode: of 1,485 grid failures, 96.5% terminate normally (semantic failures) and only 3.5% truncate, so the bit's recall is 1.0 on truncation but 0.035 overall, and on the six-provider grid it collapses to ρ=0.51 versus the gate's 0.94. Semantic failures need the paid judge, which does carry signal (ρ=0.37, AUC 0.72). This division of labor is the *calibrate-then-trust* recipe: screen routine changes with the free bit for truncation regressions, and reserve the $0.60/decision judge for the semantic failures and near-equal ranking (§4.2) it alone resolves.

4.6 Negative results and additional analyses

Recalibration does not transfer, which is why we recommend calibrate-then-trust over a gate fix. Both natural repair routes fail out-of-sample: a human-anchored persona-reweighting lifts in-sample validity from ρ=0.71 to 0.80, but the per-persona bias term does not hold under leave-one-stratum-out (dropping to ρ=0.68); combining all four judges gives no lift over the single best and is worse under leave-one-agent-out. Both gains are in-sample artifacts (Appendix F.3): the dependable path is to learn the gate's trusted operating region and operate within it (§4.5), rather than attempting to repair it. Cheap substitutes for the audit fare no better on the near-equal regime: across 21 candidate signals the four-judge average reproduces the best single judge's 31% error exactly (27/87), a common-mode failure (the judges correlate 0.67-0.98 and flip the same pairs; Appendix G); only *calibrated abstention*, declining pairs whose gate gap is within its sampling noise, reduces it, halving error on the pairs it ranks (14.8%).

5 Discussion and Guidance

A gate can be human-validated yet still mis-rank on success: ranking and construct validity are separate properties, and certifying one against human satisfaction does not guarantee the other. Framed positively, satisfaction and task completion are complementary axes (satisfaction validly captures experience), so a release gate should certify both rather than treat one as a proxy for the other. The cost-reducing remedy is a cadence (Table 3): run a cheap signal on every config change and the expensive verifiable audit on a trigger rather than a fixed calendar (a change of judge or simulator, a shift in the candidate pool, or a near-equal candidate set), to calibrate the cheap signal rather than replace it.

Three guidelines follow. **Never gate on satisfaction alone, especially when optimizing**: once the gate is an optimization target (prompt/policy/model search), tuning to a gate raises a score decorrelated from success, so pair any satisfaction signal with a verifiable check. **Aggregate to the variant**, never a single (variant$\times$persona) cell. **When judge and candidate share a provider, calibrate thresholds against an out-of-family judge**: the +0.75/7 self-inflation (§4.3) can flip an accept/reject bar even with the ranking intact.

Limitations

Human panel size. The panel comprises 3 annotators rating 150 transcripts in a fully-crossed design (Krippendorff's α=0.79, human–human ceil-

ing ρ=0.85). While sufficient for the headline satisfaction–success decorrelation ($\rho = -0.147$, replicated on SimulatorArena at 38.7%), a larger panel would tighten per-dimension confidence intervals. The headline is robust to any single rater: leave-one-annotator-out keeps it within the 56.4–60.5% band around 57.5% (Appendix A).

Degradation coverage. The controlled-degradation set uses budget-style perturbations (output-token caps, step caps, error-abort; Appendix C), so its broken-vs-working result detects truncated agents rather than every failure mode. The full capability ladder provides complementary coverage of naturalistic failures.

Synthetic substrates. τ^2-bench user instructions are synthetic, and our absolute pass-rates are not leaderboard-comparable because the harness runs on Bedrock with a tolerant output parser (Appendix K). The parser is lossless by construction, leaving the verifiable reward's deterministic DB-state and action checks unchanged, but generalization to live customer interactions remains untested.

Ethics Statement

Auditing customer-agent release gates supports safer deployment: satisfied-but-failed inversions are silent failures that harm customers without surfacing in satisfaction-based monitoring. The human study rated only de-identified, synthetic-task transcripts (no personal data) and was therefore IRB-exempt; the three consenting adult annotators (representative lay raters, independent of the project team) were compensated above the local minimum wage and rated blind to the AI scores, variant, and outcome.

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. 2018. Sanity checks for saliency maps. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, NIPS'18, page 9525–9536, Red Hook, NY, USA. Curran Associates Inc.

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. 2016. Concrete problems in AI safety. *Preprint*, arXiv:1606.06565.

Anthropic. 2025. Claude Haiku 4.5 system card. https://www.anthropic.com/claude-haiku-4-5-system-card. Official system card.

Anthropic. 2026a. Claude Opus 4.6 system card. https://www.anthropic.com/claude-opus-4-6-system-card. Official system card.

Anthropic. 2026b. Claude Opus 4.8 system card. https://www.anthropic.com/claude-opus-4-8-system-card. Official system card.

Anthropic. 2026c. Claude Sonnet 4.6 system card. https://www.anthropic.com/claude-sonnet-4-6-system-card. Official system card.

Ron Artstein and Massimo Poesio. 2008. Survey article: Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.

Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2026. τ^2-bench: Evaluating conversational agents in a dual-control environment. In *Proceedings of the 43rd International Conference on Machine Learning*.

Umesh Bodhwani, Yuan Ling, Shujing Dong, Yarong Feng, Hongfei Li, and Ayush Goyal. 2025a. A calibrated reflection approach for enhancing confidence estimation in LLMs. In *Proceedings of the 5th Workshop on Trustworthy NLP (TrustNLP 2025)*, pages 399–411, Albuquerque, New Mexico. Association for Computational Linguistics.

Umesh Bodhwani, Yuan Ling, Cibi Chakravarthy Senthilkumar, Shujing Dong, Yarong Feng, Hongfei Li, and Ayush Goyal. 2025b. LentEx: Generalizable latent entity extraction via synthetic data and instruction-tuned LLMs. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8.

Samuel R. Bowman and George Dahl. 2021. What will it take to fix benchmarking in natural language understanding? In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4843–4855, Online. Association for Computational Linguistics.

Serina Chang, Ashton Anderson, and Jake M. Hofman. 2025. ChatBench: From static benchmarks to human-AI evaluation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 26009–26038. Association for Computational Linguistics.

Lee Joseph Cronbach and Paul E. Meehl. 1955. Construct validity in psychological tests. *Psychological bulletin*, 52 4:281–302.

Sam Davidson, Salvatore Romeo, Raphael Shu, James Gung, Arshit Gupta, Saab Mansour, and Yi Zhang. 2023. User simulation with large language models for evaluating task-oriented dialogue. *Preprint*, arXiv:2309.13233.

DeepSeek-AI. 2024. DeepSeek-V3 technical report. *Preprint*, arXiv:2412.19437.

DeepSeek-AI. 2025. DeepSeek-V3.2 model card. https://huggingface.co/deepseek-ai/DeepSeek-V3.2. Official model card.

Yao Dou, Michel Galley, Baolin Peng, Chris Kedzie, Weixin Cai, Alan Ritter, Chris Quirk, Wei Xu, and Jianfeng Gao. 2025. SimulatorArena: Are user simulators reliable proxies for multi-turn evaluation of AI assistants? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 35212–35290. Association for Computational Linguistics.

Bradley Efron. 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26. DOI:10.1214/aos/1176344552.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. The Llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. 2024. A survey on LLM-as-a-Judge. *Preprint*, arXiv:2411.15594.

Abigail Z. Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. DOI:10.1145/3442188.3445901. arXiv:1912.05511.

Jaehun Jung, Faeze Brahman, and Yejin Choi. 2025. Trust or escalate: LLM judges with provable guarantees for human agreement. In *The Thirteenth International Conference on Learning Representations*.

Klaus Krippendorff. 1980. *Content Analysis: An Introduction to Its Methodology*. Sage.

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2025. RewardBench: Evaluating reward models for language modeling. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1755–1797, Albuquerque, New Mexico. Association for Computational Linguistics.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2024. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations (ICLR)*.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-eval: NLG evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522, Singapore. Association for Computational Linguistics.

Meta. 2024. Llama 3.3 70B Instruct model card. https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct. Official model card.

Mistral AI. 2025. Introducing Mistral 3. https://mistral.ai/news/mistral-3/. Official release announcement for Mistral Large 3 and the Ministral 3 family.

OpenAI. 2026a. Introducing GPT-5.4. https://openai.com/index/introducing-gpt-5-4/. Official release announcement.

OpenAI. 2026b. Introducing GPT-5.5. https://openai.com/index/introducing-gpt-5-5/. Official release announcement.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. Llm evaluators recognize and favor their own generations. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS '24, Red Hook, NY, USA. Curran Associates Inc.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, dahai li, Zhiyuan Liu, and Maosong Sun. 2024. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *The Twelfth International Conference on Learning Representations*.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics.

Ivan Sekulic, Silvia Terragni, Victor Guimarães, Nghia Khau, Bruna Guedes, Modestas Filipavicius, Andre Ferreira Manso, and Roland Mathis. 2024. Reliable LLM-based user simulator for task-oriented dialogue systems. In *Proceedings of the 1st Workshop on Simulating Conversational Intelligence in Chat (SCI-CHAT 2024)*, pages 19–35, St. Julians, Malta. Association for Computational Linguistics.

Preethi Seshadri, Samuel Cahyawijaya, Ayomide Odumakinde, Sameer Singh, and Seraphina Goldfarb-Tarrant. 2026. Lost in simulation: LLM-simulated users are unreliable proxies for human users in agentic evaluations. In *Proceedings of the 64th Annual*

Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 47423–47439. Association for Computational Linguistics.

Patrick E. Shrout and Joseph L. Fleiss. 1979. Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428. DOI:10.1037/0033-2909.86.2.420.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward hacking. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA. Curran Associates Inc.

Marilyn Strathern. 1997. 'improving ratings': audit in the british university system. *European Review*, 5:305 – 321.

Rickard Stureborg, Dimitris Alikaniotis, and Yoshi Suhara. 2024. Large language models are inconsistent and biased evaluators. *Preprint*, arXiv:2405.01724.

Weiwei Sun, Shuo Zhang, Krisztian Balog, Zhaochun Ren, Pengjie Ren, Zhumin Chen, and Maarten de Rijke. 2021. Simulating user satisfaction for the evaluation of task-oriented dialogue systems. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '21, page 2499–2506, New York, NY, USA. Association for Computing Machinery.

Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Yuan Tang, Alejandro Cuadron, Chenguang Wang, Raluca Popa, and Ion Stoica. 2025. Judgebench: A benchmark for evaluating LLM-based judges. In *The Thirteenth International Conference on Learning Representations*.

Haoxin Wang, Xianhan Peng, Huang Cheng, Yizhe Huang, Ming Gong, Chenghan Yang, Yang Liu, and Jiang Lin. 2025. ECom-bench: Can LLM agent resolve real-world E-commerce customer support issues? In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 276–284, Suzhou (China). Association for Computational Linguistics.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, Bangkok, Thailand. Association for Computational Linguistics.

Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. Self-Instruct: Aligning language models with self-generated instructions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13484–13508. Association for Computational Linguistics.

Ziyi Wang, Yuxuan Lu, Wenbo Li, Amirali Amini, Bo Sun, Yakov Bart, Weimin Lyu, Jiri Gesi, Tian Wang, Jing Huang, Yu Su, Upol Ehsan, Malihe Alikhani, Toby Jia-Jun Li, Lydia Chilton, and Dakuo Wang. 2026. OPeRA: A dataset of observation, persona, rationale, and action for evaluating LLMs on human online shopping behavior simulation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 43942–43960. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik R Narasimhan. 2025. {τ}-bench: A benchmark for \underline{T}ool-\underline{A}gent-\underline{U}ser interaction in real-world domains. In *The Thirteenth International Conference on Learning Representations*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, Red Hook, NY, USA. Curran Associates Inc.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations (ICLR)*. ArXiv:2307.13854.

Xuhui Zhou, Weiwei Sun, Qianou Ma, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, and Maarten Sap. 2026. Mind the sim2real gap in user simulation for agentic tasks. In *Conference on Language Modeling (COLM)*. ArXiv:2603.11245.

Mingchen Zhuge, Changsheng Zhao, Dylan R. Ashley, Wenyi Wang, Dmitrii Khizbullin, Yunyang Xiong, Zechun Liu, Ernie Chang, Raghuraman Krishnamoorthi, Yuandong Tian, Yangyang Shi, Vikas Chandra, and Jürgen Schmidhuber. 2025. Agent-as-a-judge: Evaluate agents with agents. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 80569–80611. PMLR.

A Satisfied-but-Failed Rates by Rater Population

Table 4 reports the satisfied-but-failed rate behind Figure 2a and the §4.1 claim: for each indepen-

Rater population	Failed/Sat.	Rate %	p
τ^2-bench customer service			
Human panel (3 raters)	23/40	57.5	2×10^{-7}
Opus-4.8 gate	30/63	47.6	8×10^{-7}
Opus-4.8 proxy	50/84	59.5	3×10^{-15}
GPT-5.5 gate	29/57	50.9	2×10^{-7}
GPT-5.5 proxy	75/126	59.5	3×10^{-22}
SimulatorArena math tutoring			
Human rating ($\geq$8/10)	12/31	38.7	0.013

Table 4: Satisfied-but-failed rate per rater population: of conversations a population rated satisfied (human/Anthropic-scale $\geq$5/7; the GPT scale, which grades $\approx$1 point lower, $\geq$4/7; SimulatorArena $\geq$8/10), the percentage the verifiable oracle scored as failed. p is a one-sided exact binomial test against the lenient 20% false-accept ceiling a construct-valid gate would meet. Every population exceeds it; the gap is not a single-rater, single-provider, or single-substrate artifact.

dent rater population, the share of conversations it rated *satisfied* that the non-LLM oracle scored as a task failure, with an exact one-sided binomial test against a lenient 20% ceiling, the false-accept rate a construct-valid *gate* should not exceed; for the satisfaction signals the primary comparator is the base failure rate (Table 2), which they sit at. Every population's satisfied-but-failed rate is far above 20%: the human panel and all four LLM scorings on τ^2 reject the ceiling at $p < 10^{-6}$, and the independent SimulatorArena human ratings at p=0.013.

Leave-one-annotator-out. The panel headline does not depend on any single rater: removing any one of the three annotators keeps the satisfied-but-failed rate within 56.4–60.5% (a 4.1-point spread around the reported 57.5%), and each annotator independently shows the same inversion (58.1%, 63.5%, 59.5% satisfied-but-failed).

B The Construct Gap Is Not Specific to Satisfaction

The main text reports *satisfaction* as the primary subjective signal because it is the construct the release gate's rubric and the LLM-proxy optimize, and the target the LLM-judge literature validates against. The blind human panel, however, rated *five* subjective dimensions per transcript (satisfaction, respect/tone, clarity, perceived helpfulness, and would-return). Table 5 shows the construct gap generalizes beyond satisfaction: every dimension shows no association with verifiable task success (all $|\rho| \leq 0.17$), and each inverts at a comparable rate (56–64% of highly-rated conversations failed). The dimensions are highly collinear (pairwise Spearman 0.61–0.97): they are not five independent tests but roughly one latent "this interaction felt good" factor, measured five ways, that is orthogonal to task success, which is why the lone nominally significant correlation (would-return, p=0.04, and *negative*) is a weak effect and, as the only nominal hit among five correlated tests, consistent with chance. Satisfaction is, if anything, the conservative choice: its inversion rate (57.5%) is among the lowest of the five. None of the other subjective dimensions would close the gap.

Human dimension	mean	ρ_{succ}	p	high-fail %	α
Satisfaction	3.88	-0.147	0.07	57.5	0.79
Respect / tone	5.26	-0.118	0.15	63.6	0.76
Clarity	5.02	$+0.085$	0.30	56.5	0.83
Perceived helpfulness	4.10	-0.129	0.12	63.3	0.80
Would return	4.19	-0.168	0.04	61.7	0.81

Table 5: Every subjective dimension the human panel rated, not satisfaction alone, is decorrelated from verifiable success (n=150). ρ_{succ} is the Spearman correlation of the 3-annotator consensus with verifiable reward; "high-fail %" is the percentage of conversations rated $\geq$5/7 on that dimension that failed the task; α is the inter-annotator Krippendorff's α (interval). Satisfaction (the gate's optimized construct, bold) is the primary case in the main text and the most conservative.

C The Controlled-Degradation Set

Purpose and design principle. The headline cross-provider grid (25 agents) establishes *external* validity (does the gate rank real, deployable models correctly?), but it cannot cleanly characterize the gate's behavior on a *known-bad* agent, because frontier models are all competent. The controlled-degradation set supplies that missing *internal* validity: a single model (Claude Sonnet-4.5) is run under 12 configurations of a-priori-known quality, so any rating difference is attributable to agent behavior rather than to provider or prompt-style confounds. This instantiates the controlled-perturbation paradigm reviewed in §2 (Ribeiro et al., 2020; Adebayo et al., 2018): the known-degradation "sanity check" that crippling a system must register on a valid metric, applied here to a release gate.

Flag-only degradation. Crucially, every configuration shares the *identical* agent prompt

and tool set; we perturb only inference-time flags: sampling temperature, the output-token cap (`max_tokens`), the conversation step cap (`max_steps`), and the orchestrator error tolerance (`max_errors`). No prompt is rewritten and no code is edited. Each degradation is a misconfiguration a real deployment could plausibly ship: a too-low output cap truncates tool-call JSON and confirmations mid-emission; a too-low step cap cuts the trajectory off before the verifiable action (τ^2 retail tasks require authenticate $\rightarrow$ locate $\rightarrow$ act $\rightarrow$ confirm); an aggressive error-abort terminates the episode on the first malformed tool call. This keeps the intervention transparent and reproducible while breaking verifiable task completion rather than surface fluency, precisely the regime in which satisfaction can diverge from success. Because these are *truncation-style* failures, the judge-free completion bit tracks them well here (ρ=0.87; §4.5); on the naturalistic six-provider grid, where 96.5% of failures instead terminate normally (semantic), the same bit collapses, which is why we scope it as a truncation-specific tripwire rather than a general regression detector.

Positive control: the ground-truth ordering is known. By construction the four `D` (degraded) configurations must rank below the eight `A`/`B` (good/medium) ones. The non-LLM verifiable reward confirms a large, clean separation (degraded-tier mean reward **0.05** versus **0.65** for good/medium), so the set behaves as a positive control: a release signal that fails to place the `D` configs last is exhibiting a measurable validity failure, not noise. Table 6 gives the full specification and every signal's per-variant mean.

C.1 Gate-score calibration

Figure 5 plots the empirical task-success rate at each integer gate score, with Wilson 95% confidence intervals, for both the controlled-degradation set and the six-provider grid. Both are increasing overall (Spearman ρ=0.86 and 0.96 over the seven score bins), so a higher gate score does on average mean a higher chance of success. Calibration is, however, cleaner on the diverse grid: on the controlled set the mid-range scores (4–5) sit below score 3 in point estimate, but those bins are small (n=54, 60) and their intervals overlap score 3's, so the local dip is not individually significant. Read conservatively, the data show only that mid-range gate scores carry little additional success signal on the degraded set; the local dip does not establish reliable non-monotonicity. We therefore treat the gate as a *ranking* instrument (§4.2) rather than a calibrated probability.

Config	Perturb.	Rew.	Gate	Prx.	Hum.
Good tier (temperature only; full budgets)					
A1 standard	T=0	0.67	4.08	4.62	4.15
A2 strict	T=0	0.63	4.20	4.63	4.15
A3 temp0.2	T=0.2	0.68	4.50	4.73	3.24
Medium tier (higher temp; mild caps)					
B3 temp0.7	T=0.7	0.68	4.42	4.72	4.67
B4 tight-budget	T0.7,tok256	0.62	4.07	4.32	4.53
B5 high-var.	T=1.0	0.65	4.17	4.87	4.25
B6 temp0.5	T=0.5	0.62	4.22	4.67	4.52
B7 tok384	T0.3,tok384	0.62	4.27	4.78	3.85
Degraded tier (budget/step/error starvation)					
D6 trunc.	tok96	0.20	2.27	3.68	2.94
D7 step-starv.	**st6**	**0.00**	**2.70**	**4.87**	**4.64**
D8 compound	T1.0,tok48,err1	0.00	1.65	3.85	2.69
D9 tok72/st12	tok72,st12	0.02	2.38	4.05	3.12

Table 6: The 12 controlled-degradation configurations (all Claude Sonnet-4.5, flag-only). Columns: verifiable reward (pass-rate), and mean LLM-judge **Gate**, LLM-**Pr**o**x**y, and **Hum**an-panel satisfaction (1–7). `tok`=`max_tokens`, `st`/`steps`=`max_steps`, `err`=`max_errors`. The degraded tier collapses on verifiable reward ($\approx$0) yet every subjective signal still rates it mid-scale. **D7** (highlighted) is the sharpest inversion: reward 0.00 but human satisfaction 4.64, the highest of any degraded config, because a step-capped trajectory is cut off mid-task while reading as helpful turn-by-turn.

D Second-Provider Simulator Replication

With a controlled swap, we test directly whether the satisfied-but-failed gap is an artifact of one over-cooperative simulator rather than a property of the satisfaction signal. Holding *everything else fixed* (the same 12 degradation variants, 6 persona strata, and identical task ids, the same rubrics and judge, and the same non-LLM oracle), we re-drive the grid with the user-simulator changed to a different provider, OpenAI GPT-5.4. The simulator provider is then the only variable that differs from the matched arm, so any persistence of the gap cannot be a single-simulator effect.

The inversion persists under the swap (Table 7). Under *both* simulators the satisfied-but-failed rate sits far above the 20% a construct-valid signal would target: GPT-5.4 66.7% (gate) and 69.8% (proxy) versus Sonnet-4.5 43.6% and 52.9% on the matched cells, all with $p < 10^{-7}$. We read this as a qualitative invariant (satisfaction fails to

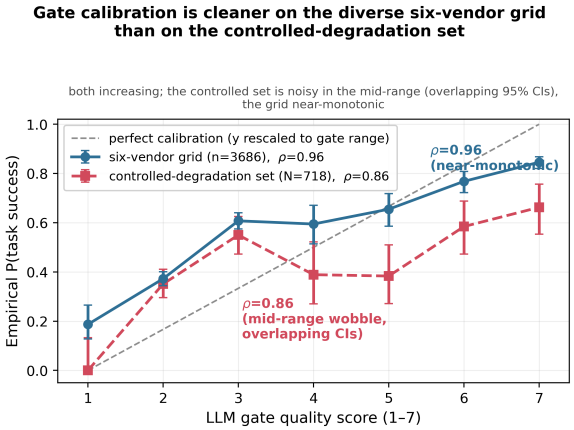


Figure 5: Gate-score calibration with Wilson 95% confidence intervals: empirical $P(\text{success})$ at each gate score on the controlled-degradation set (N=718 gate-scored, Spearman ρ=0.86) and the six-provider grid (n=3,686 gate-scored, ρ=0.96). The grid is near-monotonic; the controlled set dips in the mid-range (scores 4–5 below score 3), but with small bins and overlapping intervals the dip is not significant. Calibration quality depends on the agent population.

track success under an independent simulator), *not* as a rate comparison: the two simulators steer the same agent into different trajectories and so yield different mean verifiable reward (GPT-5.4 0.21 vs. Sonnet 0.32, the harsher simulator surfacing more task failures), which mechanically shifts the absolute false-accept rate. The defensible claim is therefore that the gap is large and highly significant under both simulator families.

Full ranking grid under GPT-5.4. The swap above isolates the *construct gap*; we also re-ran the entire 25-agent *ranking* grid with the GPT-5.4 user-simulator (114 tasks per agent, 2,850 transcripts), holding agent, tasks, rubrics, judge, and oracle fixed. The agent ordering is preserved: Spearman ρ=0.93 (Pearson 0.97; base-model cluster-bootstrap 95% CI $[0.71, 0.98]$), against 0.94 under Sonnet-4.5, and the near-equal degradation reproduces (top-11 ρ=0.51). Same-family simulator–agent affinity therefore does not account for the ranking result (§4.2).

E Per-Agent Data Behind Figure 4

Table 8 gives the full per-agent values plotted in Fig. 4 (the cross-provider dumbbell): for each of the 25 agents, the verifiable reward (mean $\pm$ SE), the LLM-judge gate and LLM-proxy satisfaction (1–7 means), the *satisfaction surplus* (proxy mapped to $[0, 1]$ minus reward), and each agent's

User-simulator	Mean rew.	Gate failed %	Proxy failed %
Sonnet-4.5 (Anthropic)	0.32	43.6	52.9
GPT-5.4 (OpenAI)	0.21	66.7	69.8

Table 7: Controlled second-provider simulator swap on the retail degradation grid: identical 12 variants $\times$ 6 strata $\times$ task ids (n=216 each), same Sonnet-4.5 agent, only the user-simulator provider differs. The satisfied-but-failed (false-accept) rate stays far above the 20% null under both simulators, so it is not a Sonnet-simulator artifact. "Satisfied" is $\geq$5/7; "failed" is verifiable reward $= 0$. The two simulators yield different mean agent reward (GPT-5.4 harsher), so compare each rate to the null, not to each other.

rank (1=best) under the verifiable reward and under the two frontier judges Opus-4.8 and GPT-5.5. Agents are grouped by provider with providers ordered by mean reward (low$\rightarrow$high capability), matching the figure's top-to-bottom layout. The surplus shrinks across the capability tiers (**surplus vs. reward Spearman** $\rho = \mathbf{-0.51}$, $p = 8.5\text{e}{-3}$, n=25): large and positive for Mistral and Qwen ($+0.20$ to $+0.40$), mixed for Meta (-0.06 to $+0.20$), and near zero or negative for the frontier (DeepSeek, OpenAI, and the strongest Anthropic configs). The rank columns make the judges' agreements and disagreements explicit: both frontier judges place the OpenAI agents in their top three on *satisfaction*, a same-style preference the verifiable reward does not share, as it ranks the GPT-5.4 configs at 7–9 and places Anthropic's Opus-4.6 first on ground-truth task success.

F Per-Judge Ranking Validity Across Providers and Domains

Section 4.2 summarizes ranking validity; Fig. 6 shows the underlying per-judge gate-vs-reward scatters in full. On the τ^2 grid the gate's ranking is recovered by all four judges, the primary Opus-4.8 (ρ=0.94) and GPT-5.5 (0.84), and the deployed Sonnet-4.5 (0.92) and GPT-5.4 (0.94); the two frontier judges' domain splits are Opus-4.8 retail 0.95, airline 0.80, and GPT-5.5 retail 0.87, airline 0.65.

F.1 The near-equal regime persists under frontier judges

We test whether the judge is weaker than the strongest agents it scores, which would make the near-equal top-11 ceiling (§4.2) an artifact of judge capability rather than a real property of the agents.

Agent	Reward (mean ± SE)	Gate	Proxy	Surplus	Rank (1=best) by Reward	Opus-4.8	GPT-5.5
Meta ($\bar{r} = 0.34$)							
Llama-3.1-8B (t.7)	0.32±0.06	2.16	4.13	+0.20	23	25	25
Llama-3.3-70B (t.7)	0.33±0.06	2.32	2.76	−0.04	22	22	18
Llama-3.1-8B	0.35±0.06	2.08	3.79	+0.12	19	24	24
Llama-3.3-70B	0.35±0.06	2.18	2.71	−0.06	20	21	21
Mistral ($\bar{r} = 0.40$)							
Mistral-Small	0.28±0.04	1.93	3.99	+0.22	25	23	23
Mistral-Small (t.7)	0.30±0.04	2.16	4.01	+0.20	24	20	22
Ministral-3B (t.7)	0.35±0.04	2.68	4.45	+0.23	21	19	20
Ministral-3B	0.44±0.04	2.69	4.83	+0.20	17	17	19
Mistral-Large-3 (t.7)	0.51±0.04	4.24	5.62	+0.26	15	15	14
Mistral-Large-3	0.54±0.04	4.40	5.63	+0.24	14	14	15
Qwen ($\bar{r} = 0.49$)							
Qwen3-32B (t.7)	0.37±0.04	2.54	5.60	+0.40	18	16	16
Qwen3-32B	0.44±0.04	2.66	5.45	+0.30	16	18	17
Qwen3-235B	0.57±0.04	4.16	5.68	+0.21	13	13	6
Qwen3-235B (t.7)	0.59±0.04	4.23	5.80	+0.21	12	12	4
DeepSeek ($\bar{r} = 0.77$)							
DeepSeek-V3.2 (t.7)	0.75±0.03	5.09	5.53	+0.01	8	9	5
DeepSeek-V3.2	0.78±0.03	5.20	5.62	−0.01	6	5	7
OpenAI ($\bar{r} = 0.78$)							
GPT-5.4 (t.7)	0.74±0.03	5.51	5.57	+0.02	9	3	3
GPT-5.4	0.77±0.03	5.62	5.56	−0.01	7	2	2
GPT-5.5	0.84±0.03	5.54	5.79	−0.04	3	1	1
Anthropic ($\bar{r} = 0.78$)							
Haiku-4.5 (t.7)	0.66±0.04	4.68	5.77	+0.13	11	10	13
Haiku-4.5	0.68±0.04	4.71	5.85	+0.12	10	11	12
Sonnet-4.6 (t.7)	0.80±0.03	4.78	5.92	+0.02	5	7	11
Sonnet-4.6	0.81±0.03	4.96	5.89	+0.00	4	8	10
Opus-4.6	0.84±0.03	5.03	5.97	−0.01	2	6	8
Opus-4.6 (t.7)	0.87±0.03	5.00	5.97	−0.04	1	4	9

Table 8: Per-agent data behind Fig. 4, all 25 agents. Reward is the verifiable pass-rate (mean ± standard error across the agent's transcripts); Gate and Proxy are 1–7 satisfaction means; Surplus = (proxy−1)/6− reward, i.e. how far satisfaction overstates verifiable success on a common $[0, 1]$ scale. The last three columns give each agent's rank (1=best) by verifiable reward and by the two frontier judges Opus-4.8 and GPT-5.5. Providers are ordered by mean reward $\bar{r}$. Surplus vs. reward: Spearman $\rho = -0.51$ ($p = 8.5\text{e}{-3}$). The rank columns expose where judges depart from ground truth: both frontier judges rank the GPT agents 1–3 on satisfaction, well above their reward ranks (GPT-5.4 at 7–9), a same-style preference, whereas the verifiable reward ranks Anthropic's Opus-4.6 first; the judges still track the overall ordering (full-ladder $\rho \geq 0.84$, §4.3).

Figure 7 rules this out. It places, for all four judges, the gate-vs-reward Spearman ρ on the full 25-agent ladder beside the same ρ on the near-equal top-11 subset. Two patterns stand out: every judge recovers the full ladder (ρ=0.84–0.94), and on the top-11 no judge recovers the ordering (ρ=0.25–0.51, all n.s. at n=11). The two frontier judges (Opus-4.8, GPT-5.5), which match the strongest agents in the pool by verifiable pass-rate (§4.3), do no better there than the deployed Sonnet-4.5 judge. If the ceiling were a judge-capability artifact, the most capable judges would escape it. The near-equal limit therefore reflects agent near-equality bounded by oracle resolution, not judge capability.

Decision-disagreement rate (§4.2). Table 9 gives, per signal, the fraction of agent pairs on which the gate promotes the lower-reward agent, split by whether the pair's verifiable rewards are near-equal ($|\Delta R| < 0.1$, n=87 pairs) or wide ($|\Delta R| \geq 0.1$, n=211). Every signal is near-perfect on wide pairs and degrades sharply on close ones; the effect is not sensitive to the exact 0.1 threshold (Opus-4.8: 45%, 31%, 26% at $|\Delta R| < 0.05/0.10/0.15$).

Is the 31% reference-ranking noise or an artifact of non-independent pairs? Two checks rule this out. A base-model cluster bootstrap (13 clusters, resampling base models rather than pairs)

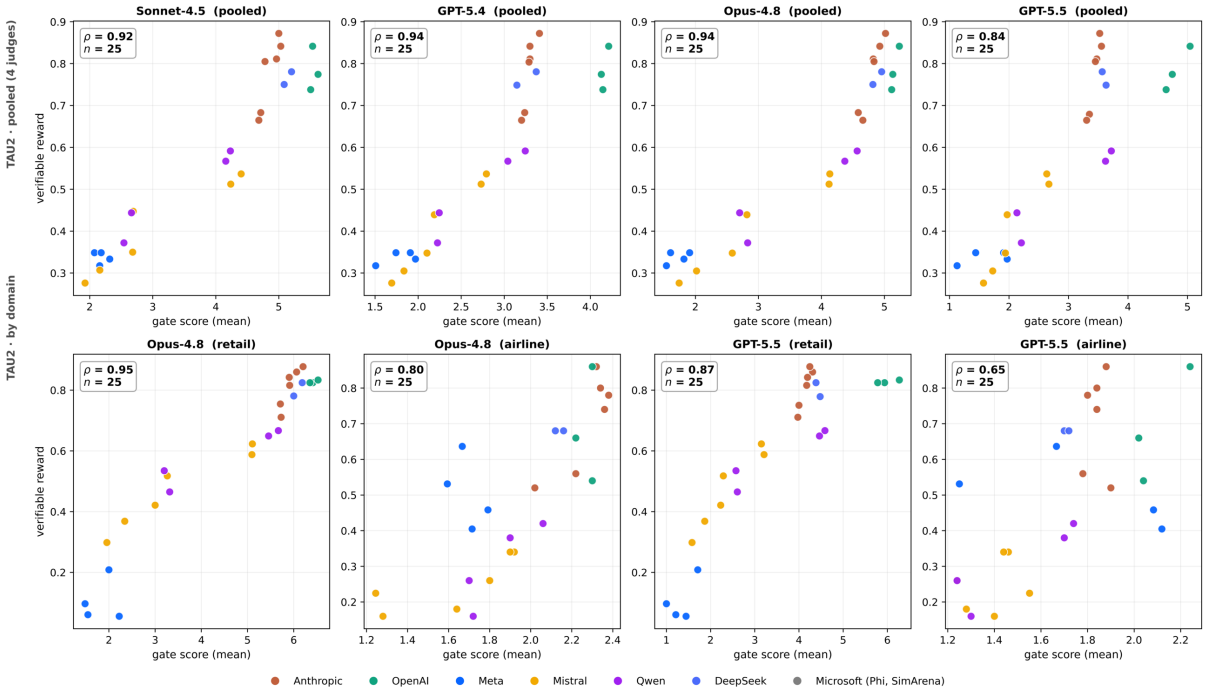


Figure 6: Per-judge gate-vs-reward validity across all available judges and domains. τ^2 rows: four judges pooled (Sonnet-4.5, GPT-5.4, Opus-4.8, GPT-5.5) and the Opus-4.8/GPT-5.5 retail vs airline splits; each point is an agent, provider-colored, with Spearman ρ annotated.

Signal	Near-equal %	Wide %
Opus-4.8 gate (primary frontier)	31.0	0.9
GPT-5.4 gate	29.9	1.4
GPT-5.5 gate	39.1	7.6
Pooled gate (`gate_quality`)	37.9	1.4
LLM-proxy satisfaction	29.9	10.4

Table 9: Decision-disagreement rate by signal: fraction of pairs where the gate promotes the *lower*-reward agent, on near-equal ($|\Delta R| < 0.1$, n=87) vs. wide ($|\Delta R| \geq 0.1$, n=211) pairs; the 2 exact-tie pairs of the $\binom{25}{2}$=300 are excluded. Base-model cluster-bootstrap 95% CI on the Opus-4.8 near-equal rate: $[11.6, 50.0]$.

puts the primary Opus-4.8 near-equal rate at 31.0%, 95% CI $[11.6, 50.0]$, its lower bound still $\sim$13$\times$ the 0.9% wide-pair rate; and dropping the least-independent same-base-model temperature pairs leaves 32.0% on the 75 cross-base-model pairs (24/75). The disagreement is also signal-*specific* rather than shared ordering noise: of the 60 near-equal pairs flipped by at least one of the five signals, only 4 are flipped by all five and 22 by exactly one (mean pairwise Jaccard 0.40), so the 31% is a population-level, signal-specific property, not shared reference-ranking noise.

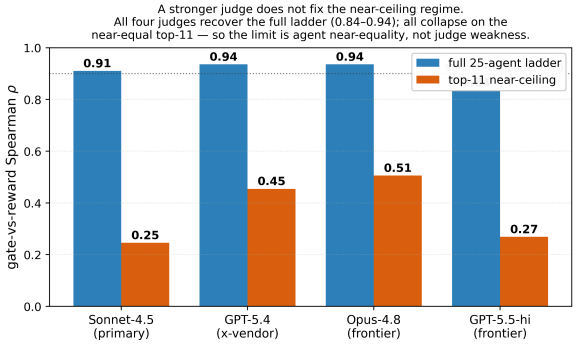


Figure 7: Four-judge robustness. All four judges, including the frontier Opus-4.8 and GPT-5.5, which match the strongest agents in the pool by verifiable pass-rate, recover the full 25-agent ladder (ρ=0.84–0.94), but none recovers the near-equal top-11 (ρ=0.25–0.51, all n.s.). The near-ceiling regime persists even for the most capable judges, so the limit reflects agent near-equality bounded by oracle resolution, not judge capability.

F.2 Same-family self-preference

The two-provider design isolates same-family self-preference as a difference of differences (§4.3): the Anthropic Opus-4.8 judge rates Claude agents above the GPT-5.5 judge, and by a *larger* margin than it rates non-Claude agents (Fig. 8). The diff-in-diff is $+0.75$/7 (the deployed Sonnet-4.5 judge shows the same effect at $+0.67$/7 against

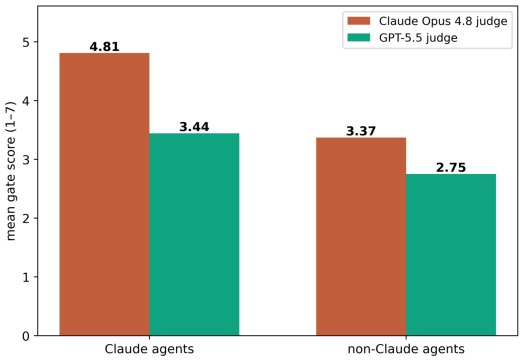


Figure 8: Same-family self-preference, shown explicitly. The Opus-4.8 judge inflates Claude agents relative to the GPT-5.5 judge more than it inflates non-Claude agents (diff-in-diff $+0.75$/7).

GPT-5.4); being scale-invariant, it reflects same-family self-preference rather than absolute harshness. Crucially, the *ranking* agreement is unchanged (ρ=0.92; §4.3), so self-preference is a measured caveat on the absolute scores while ranking validity holds.

F.3 Judge stability

Two reruns confirm the agent ranking is not an artifact of judge noise. First, in a test–retest, re-scoring 48 transcripts three times (judge temperature 0.7) yields a judge self-reliability of ICC=0.87 (Shrout and Fleiss, 1979) and a variant-level rank stability of ρ=0.92 across re-scorings, a judge-noise measure distinct from the human-panel inter-annotator agreement (Krippendorff's α=0.79). Second, running the GPT judge five independent times agrees with itself on the agent ranking at mean ρ=0.74 (min 0.52); per-run validity against the verifiable reward is noisier (ρ=0.24–0.59 across runs), which is why we report that judge as a ranking-agreement check (§4.3) rather than a per-run validity estimate.

G Cheap Signals for the Near-Equal Regime

We test whether any cheaper signal removes the near-equal decision error of §4.2 without the paid audit. Across 21 candidate signals, spanning single judges, the process-blind proxy, the completion bit, and unweighted and confidence-weighted judge ensembles, none outperforms the best single judge on near-equal pairs. Averaging all four judges gives exactly 31.0% (27/87), identical to the best single judge, because the judges are highly correlated (pairwise ρ 0.67–0.98) and flip the same close pairs: the error is *common-mode*, so ensembling cannot average it away. The one effective lever is *calibrated abstention*, declining to rank a pair whose gate-score gap falls inside its bootstrap sampling noise. On the pairs it does rank (31–49% of near-equal pairs, depending on the abstention threshold) it roughly halves the error to 14.8%, buying selectivity rather than resolution, which confirms that the limit is oracle resolution, not judge choice.

H Why Airline Ranking Is Weaker: Gate-Score Range Restriction

Range restriction, rather than a deficiency of the gate, explains the lower airline ranking validity (§4.2). Fig. 9 shows the gate-score distributions per domain: the airline gate spans a $\approx$2.2$\times$ narrower band than retail (Sonnet gate range 2.03 vs 4.56; GPT-5.4 gate 1.19 vs 3.56, a 3.0$\times$ compression), while the verifiable-reward spread is nearly identical across domains (range ratio 1.17$\times$). With agents compressed into a narrow gate band there is less to rank, which attenuates a rank correlation; both judges exhibit the same compression, confirming it is a property of the airline substrate rather than of either judge.

I Persona Strata

The user-simulator is conditioned on one of six persona strata, each a 2–4 sentence second-person behavioral overlay prepended to the τ^2-bench task instructions (the task facts, namely order IDs, customer identity, and item specs, are held fixed, so only tone, cooperativeness, and pacing change; the verifiable-reward signal is unaffected). The strata are hand-authored archetypes whose axes (cooperativeness, patience, assertiveness) are informed by the OPeRA persona schema (Wang et al., 2026) (demographics, Big-Five, and Consumer-Styles-Inventory shopping styles); we do not sample individual OPeRA records. Table 10 lists them. Persona conditioning is behaviorally real: on an identical retail task, the cooperative user writes "*Yes, I confirm. Please go ahead… I appreciate your help*" while the impatient user writes "*I don't have time for this… just do it already*" and the skeptical negotiator probes "*this 'no modification fee' sounds too good to be true, what's the catch?*"

Verbatim persona overlays. Each stratum is a second-person behavioral overlay prepended to the task instructions with the wrapper "`PERSONA: adopt this style for`

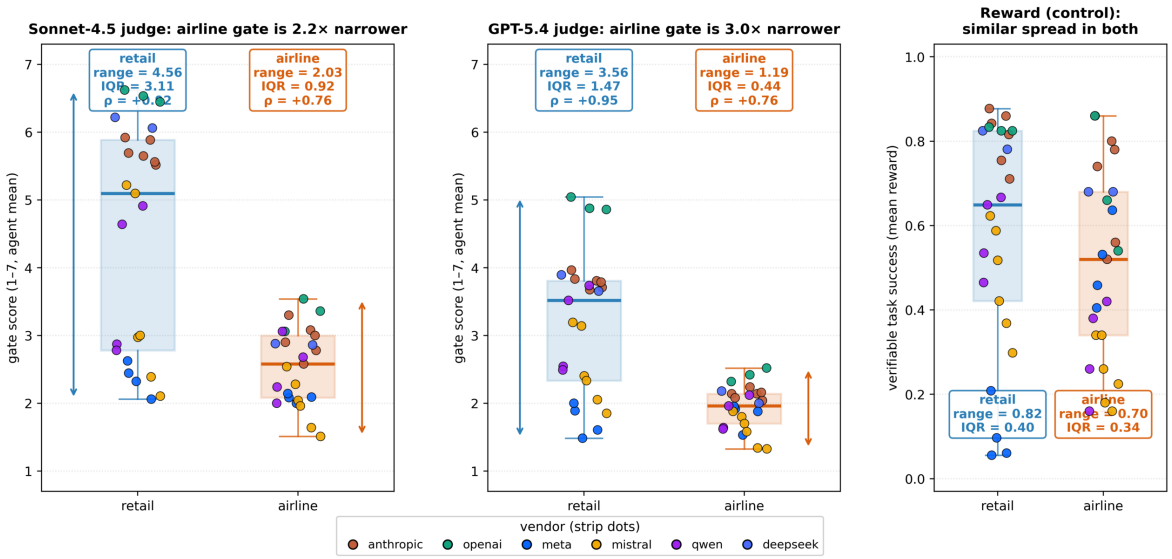


Figure 9: Gate-score distributions by domain for both judges (box + per-agent strip), with the reward distribution as a control. The airline gate signal is ≈2.2–3.0× more compressed than retail (Sonnet and GPT-5.4 alike), whereas verifiable reward is comparably spread in both domains, so the weaker airline ranking ρ is driven by gate-score range restriction, not by the gate being less valid there.

Stratum	Coop.	Patience	Assert.
S1 Cooperative	high	high	low
S2 Impatient/rude	low	low	high
S3 Distracted	med	med	low
S4 Anxious/low-trust	med	med	med
S5 Terse	high	med	med
S8 Skeptical negotiator	low	med	high

Table 10: The six persona strata and their design axes (cooperativeness, patience, assertiveness), informed by the OPeRA schema (Wang et al., 2026). The six are drawn from an eight-archetype pool; S6 and S7 (chatty over-sharer, indecisive perfectionist) are defined but not exercised in the reported runs, hence the index gap. Used in the controlled-degradation set and the 150-transcript human panel; see §4.4 for their effect on agent difficulty and ranking.

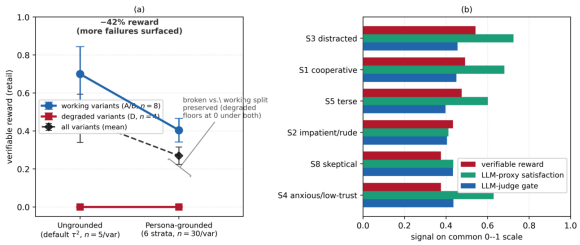


Figure 10: Effect of persona strata on agent difficulty and ranking (retail), discussed in §4.4. **(a)** Per-variant verifiable reward under the ungrounded default-τ^2 user versus the six persona strata: personas cut mean verifiable reward 42% (0.47 → 0.27) while the agent ordering holds (reward-rank ρ=0.83). **(b)** Per-stratum reward, gate, and LLM-proxy satisfaction (grounded 720-set, common $[0, 1]$ scale): satisfaction varies sharply by persona but does not track reward.

`the entire conversation; keep all facts of your request unchanged: {overlay}`", followed by the unchanged τ^2-bench task. The full overlays are:

S1 Cooperative. A courteous, even-tempered customer in their mid-30s who shops online regularly and trusts the representative. Answers directly and completely, stays focused, stays patient through multi-step resolution, confirms details without rushing, and expresses mild appreciation when things go well.

S2 Impatient/rude. A busy, short-tempered customer who feels they have wasted too much time. Blunt to the point of rudeness, interrupts with demands rather than answering procedural questions, becomes openly irritated at requests for information they think the agent should already have, pressures the agent to skip steps, and demands escalation when anything takes more than a couple of exchanges.

S3 Distracted. A multitasking customer only half paying attention, replying while distracted. Gives incomplete or slightly-off answers, misses what was just asked, and drifts onto tangents before returning; not hostile, just scattered ("sorry, what did you ask?"), so the

agent must keep them on track.

S4 Anxious/low-trust. A risk-averse customer who distrusts remote support and worries something will go wrong. Asks the agent to justify each action before agreeing, seeks repeated reassurance about charges, and hesitates before sharing details; polite but cautious ("are you sure?", "what happens if this doesn't work?").

S5 Terse. An experienced, no-nonsense customer who values speed. Responds in clipped, minimal phrases, skips pleasantries, and provides only the specific information requested; cooperative in substance but cold in style, nudging the agent to get to the point.

S8 Skeptical negotiator. A price-sensitive, deal-driven customer who treats every interaction as a negotiation and is skeptical of any policy that does not favor them. Assertively challenges fees and refusals, presses for waivers or exceptions, and probes for loopholes; persistent and adversarial but not abusive, relenting only once a limit is firmly justified.

The persona modifies only tone, cooperativeness, pacing, and assertiveness; task facts (`reason_for_call`, `known_info`, `evaluation_criteria`) are untouched, so the verifiable reward is unaffected. The two defined-but-unexercised strata (S6 chatty over-sharer, S7 indecisive perfectionist) account for the gap between S5 and S8.

J Dataset Statistics

Table 11 summarizes the three evaluation substrates. The cross-provider ladder is the headline grid; the controlled-degradation set is the positive control (§4.5); the human panel and SimulatorArena provide the human-grounded checks (§4.1). The controlled-degradation set and the persona-simulated conversations are generated synthetically; synthetic generation is a common remedy for scarce labeled data in NLP (Wang et al., 2023; Bodhwani et al., 2025b). The ladder's 14 base models are Anthropic Opus-4.6/Sonnet-4.6/Haiku-4.5 (Anthropic, 2026a,c, 2025), Meta Llama 405B/70B/8B (Grattafiori et al., 2024; Meta, 2024), Mistral Large-3/Small/Ministral-3B (Mistral AI, 2025), Qwen3 235B/32B (Yang et al., 2025), DeepSeek-V3.2 (DeepSeek-AI, 2024, 2025), and OpenAI GPT-5.4/5.5 (OpenAI, 2026a,b), each run at two temperatures.

Cross-provider ladder (τ^2-bench retail + airline)	
Scorable transcripts (non-null reward)	3,691
retail / airline	2,487 / 1,204
Agent configurations scored	25
Task pool (retail / airline)	114 / 50
Controlled-degradation set (Sonnet-4.5, flag-only)	
Transcripts	720
Cells (12 configs $\times$ 6 strata $\times$ 2 domains)	144
Transcripts per cell	5
Tiers (good / medium / degraded)	3 / 5 / 4 configs
Human panel (τ^2 subsample)	
Transcripts / annotators	150 / 3
Stratification (retail / airline)	85 / 65
failed / succeeded	86 / 64
Krippendorff's α (satisfaction)	0.79
SimulatorArena (math tutoring)	
Models / graded conversations	9 / 50
High-satisfaction subset ($\geq$8/10)	31

Table 11: Dataset statistics across the three substrates. The 25 scored agent configurations are the 28 (model, temperature) cells minus three excluded for infrastructure reasons (Llama-3.1-405B at both temperatures, throttle-limited; and the GPT-5.5 temperature-0.7 cell, as GPT-5.5 fixes its sampling setting). Conversations average 31 messages, 8.6 tool calls, and 7.6 user turns.

K Oracle Patch and Validation

The verifiable reward is the ground truth the audit rests on, so we document our only modification to the τ^2-bench evaluator, a bug-fix rather than a degree of freedom, in two disjoint parts.

(A) Backend swap (no scoring change). Four `DEFAULT_LLM_*` constants are repointed from `gpt-4.1` to `bedrock/...claude-sonnet-4-5` (temperatures stay at 0), so the agent, simulator, and LLM sub-evaluator run on Bedrock; no scoring logic is touched.

(B) Tolerant verdict parser. A `loads_tolerant` helper replaces a bare `json.loads` at two sites (tool-output comparison and NL-assertion parsing): it tries strict `json.loads` *first* and only on failure strips ```` ```json ```` fences or extracts the outermost JSON object, re-raising if nothing parses. It never rewrites or flips a parsed verdict and adds no pass condition; it only recovers verdicts strict parsing would drop because Claude wraps JSON in prose (absent in GPT JSON-mode). A 16-case unit

test confirms identical parsing on valid JSON, recovery on fenced/wrapped JSON, and a raise on unparseable input.

Why the reward is robust to the patch. The reward is the product of each task's `reward_basis` components; the DB-state, action, environment-assertion, and `communicate` checks are deterministic Python, and the NL-assertion judge is the only LLM the parser touches. Airline (`reward_basis` = (DB, COMMUNICATE)) is thus fully deterministic and patch-independent; for retail ((DB, NL_ASSERTION)) the NL term is multiplicatively gated by the deterministic DB check and changed the outcome in only 113 of 2,537 applicable simulations (4.5%). The harness persists only the parsed verdict, so we rely on this lossless-by-construction argument rather than an empirical recovery rate, and make no claim of human–oracle agreement (the panel rated experience, not task success). Crucially, the satisfaction–success gap does not depend on this component: on *airline*, whose oracle contains no LLM, the process-blind proxy still false-accepts 62.1% of satisfied conversations ($325/523$).

L Prompts and Rubrics

We reproduce the operative prompts verbatim. Model identifiers and tool names are kept as run. All scoring runs offline through a Bedrock-Converse harness; a tolerant judge-output parser recovers the score line from formatting variation, cutting null judge scores from 36.6% to near zero.

Release-gate judge (policy-aware). System prompt (abridged): *"You are a release-gate quality auditor for an automated retail customer-service agent… you evaluate ONLY the assistant, from the standpoint of an operations supervisor deciding whether this agent variant is safe to ship."* It scores four dimensions in priority order, (1) policy adherence (authenticate before acting; explicit confirmation before any database write; one tool call per turn; no fabrication), (2) task resolution, (3) efficiency, and (4) professionalism, on a 1–7 scale, explicitly instructing that "a polite agent that skipped authentication or wrote to the database without confirmation must score low," and emitting a final `SCORE: <1-7>` line.

Human-proxy (satisfaction-only). System prompt (abridged): *"You are role-playing a real online shopper who just finished a chat… answer the way an ordinary, non-expert customer would, based purely on how the conversation FELT to you, not on any knowledge of company policies, internal tools, or what the agent did behind the scenes."* It rates satisfaction, respect/tone, clarity, perceived helpfulness, and would-return on 1–7, and is shown only the user-visible spoken turns (tool calls and the task stripped out), enforcing process-blindness. This disjointness in role, evidence, and vocabulary is what prevents the gate and proxy from agreeing by shared-rubric artifact.

Human panel. The three annotators received written instructions to rate *the feeling of being served, not the mechanics*: "An agent can feel great yet not actually solve your problem, and vice versa; rate your experience." They scored the same five dimensions (1–7) plus a one-phrase "biggest problem" field, blind to the verifiable reward, variant, and model identity, on the user-visible transcript only.

User simulator. The persona-driven simulator extends the base τ^2-bench guidelines ("generate one message at a time… disclose information progressively… never hallucinate information not in the scenario… emit `###STOP###` when the goal is satisfied") with the persona overlay (Appendix I); task facts are unchanged.

M Representative Examples

One real, de-identified example per substrate (public-benchmark tasks; no customer data).

τ^2 retail (satisfied but failed). Task: a customer (Yusuf Rossi, zip 19122) asks how many t-shirt options exist and to change all pending t-shirt orders to purple, size S, v-neck, polyester. The agent authenticates by name+zip, lists the options, and reports "*Both modifications have been completed successfully.*" The user ends satisfied (`###STOP###`). **Gate 7/7, proxy 7/7, human panel 5.5/7, verifiable reward 0.0**: all three satisfaction signals passed, yet a pending order was never modified (12 of 13 tool calls correct), so the task failed despite a flawless-feeling interaction.

τ^2 airline (satisfied but failed). Task: a customer (Mohamed Silva) asks for the summed gift-card and certificate balances and to rebook a reservation to the cheapest business round-trip without changing dates. The agent reports balances and confirms the charges; the user replies "*That's ev-*

erything I needed. Thank you so much!" **Proxy 7/7, reward 0.0**: the rebooking did not match the required end-state, and the cross-provider judge (GPT-5.4 gate 1/7) flags it.

SimulatorArena math tutoring (satisfied but wrong). Problem: 11 players each pass to every other player three times; how many passes? The tutor walks through "one player makes $10 \times 3 = 30$ passes," then states "the correct total is indeed 165. Well done!" **Human rating 10/10, proxy 7/7, but verifiably incorrect**: the tutor's own shown work (30 passes per player, 11 players) implies 330, not the stated 165; the policy-aware gate (Opus-4.8 1/7) catches the error the human did not.